

Regional Growth Analysis

Which region is really growing, and can we trust the growth rate?

Veda Technology | Data Analytics Track | Project report by Sneha | Tools: SQL (SQLite) and Excel | Data: Superstore, 2014-2017

\$484k to \$733k company sales, 2014 to 2017	14.8% CAGR West 19.1%, East 18.3%, Central 12.3%, South 5.8%	+2,072% Central profit growth in 2015, on a base of only \$540
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1. Objective

The task was to compare growth rates of the four Superstore regions over time and practise period-over-period comparison. I wanted to go one step further than just calculating percentages: for each number I asked whether it is a fair picture of the region, or whether the base year, one big order or the Q4 season is playing tricks. The deliverables are a growth table, a chart and five insights.

2. Data and tools

- **Dataset:** Superstore, 9,994 order lines, orders placed between 3 Jan 2014 and 30 Dec 2017, four regions (Central, East, South, West). No missing values and no duplicate Row IDs.
- **SQL (SQLite):** built one clean view with year and quarter, then wrote eight queries using GROUP BY, CTEs and the LAG() window function (file *regional_growth_analysis.sql*).
- **Excel:** rebuilt the same tables with live SUMIFS formulas on the raw data, added conditional checks and native charts. A separate sheet compares SQL output with Excel results; all 16 region-year rows match.

3. Approach and definitions

- **Consistent periods:** calendar years (Jan-Dec) for annual growth, and each calendar quarter is compared with the *same quarter of the previous year*. Every region has all 4 years and all 16 quarters, so no partial periods.
- **YoY growth** = (this year / last year) - 1. **CAGR** = $(2017 / 2014)^{1/3} - 1$, which smooths out a bumpy year.
- **Absolute gain** in dollars is always shown next to the percentage, because a percentage alone hides size.
- **Small-base flag:** a growth number is marked SMALL BASE when the previous-year value was below 50% of the region's own 4-year average (the 50% is an input cell in Excel and can be changed).

4. Growth table

Table 1 - Sales by region (USD), CAGR and dollar gain

Region	2014	2015	2016	2017	CAGR	Gain 14-17
Central	103,838	102,874	147,429	147,098	12.3%	43,260
East	128,680	156,332	180,686	213,083	18.3%	84,402
South	103,846	71,360	93,610	122,906	5.8%	19,060
West	147,883	139,966	187,480	250,128	19.1%	102,245
Total	484,247	470,533	609,206	733,215	14.8%	248,968

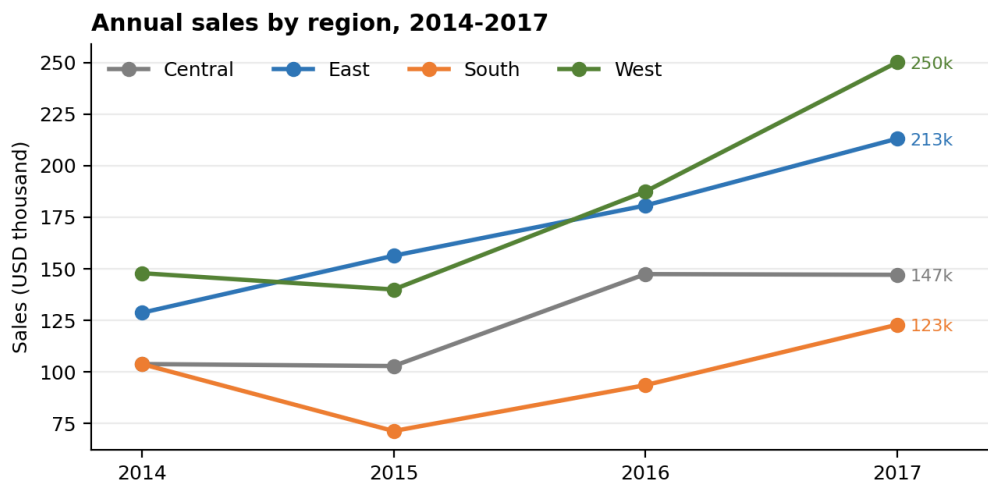


Table 2 - Year-over-year growth in sales and profit

Region	2015	2016	2017	Profit 2015	Profit 2016	Profit 2017
Central	-0.9%	+43.3%	-0.2%	+2071.6%	+69.8%	-62.0%
East	+21.5%	+15.6%	+17.9%	+23.6%	-4.5%	+65.0%
South	-31.3%	+31.2%	+31.3%	-30.0%	+112.8%	-50.0%
West	-5.3%	+34.0%	+33.4%	+2.1%	+17.4%	+82.2%
Total	-2.8%	+29.5%	+20.4%	+24.4%	+32.7%	+14.2%

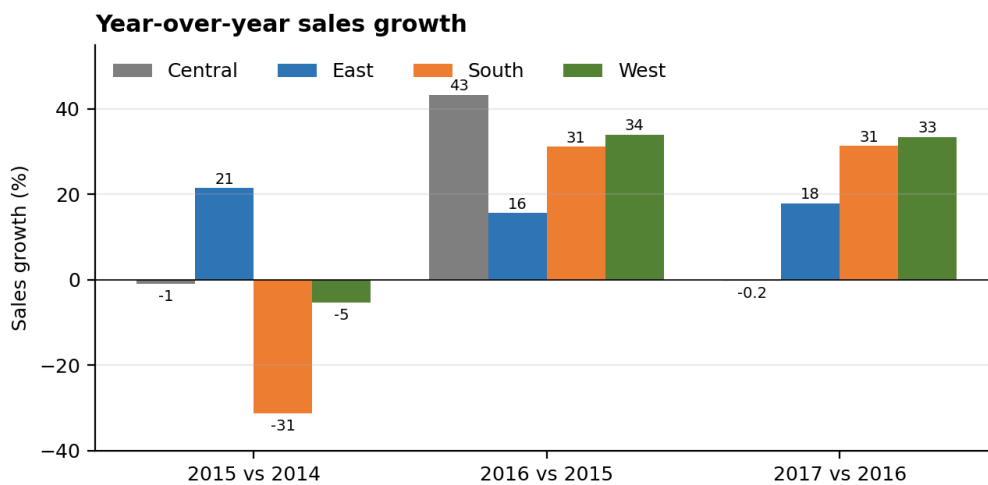


Table 3 - Profit margin (profit / sales)

Region	2014	2015	2016	2017
Central	0.5%	11.4%	13.5%	5.1%
East	13.3%	13.5%	11.2%	15.6%
South	11.4%	11.7%	18.9%	7.2%
West	13.6%	14.6%	12.8%	17.5%
Total	10.2%	13.1%	13.4%	12.7%

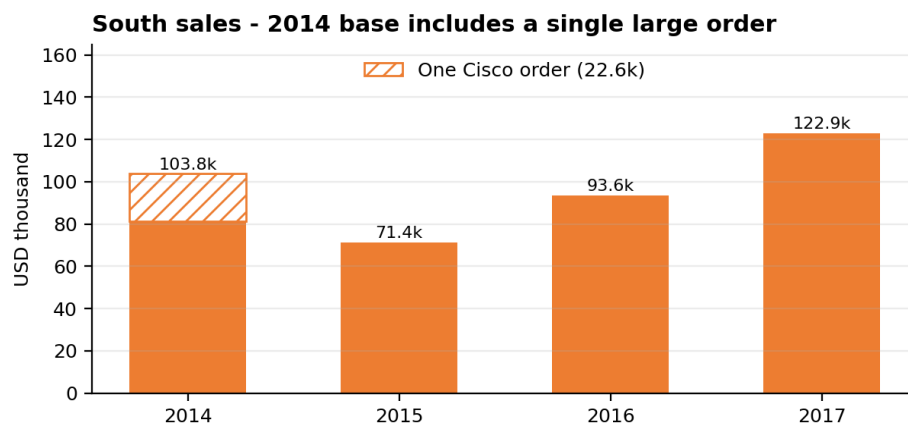
5. Five insights

Insight 1 - West and East carry the growth

West (19.1% CAGR) and East (18.3%) together added about 75% of the \$249k sales gain between 2014 and 2017 (West 41.1%, East 33.9%). East grew in every year (+21.5%, +15.6%, +17.9%), which makes it the steadier of the two. West fell 5.4% in 2015 and then grew 34% and 33%. **So what:** targets and inventory planning should lean on these two regions.

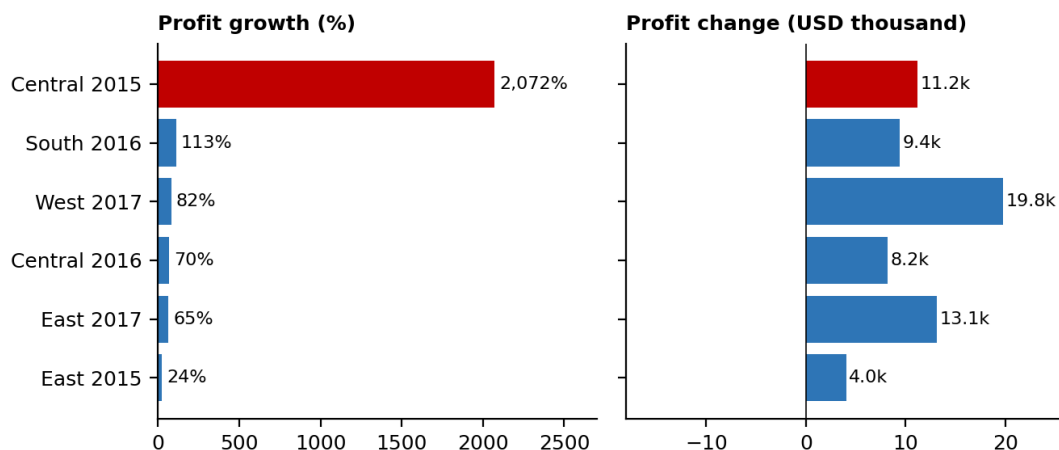
Insight 2 - South's swings are partly one big order

South looks dramatic: -31.3% in 2015, then +31.2% and +31.3%. But one order in March 2014, a Cisco TelePresence system worth \$22,638, made up 21.8% of South's 2014 sales. Without it the 2014 base is about \$81.2k and the 2015 fall is -12.1%, not -31.3%. Even with two +31% years South has the lowest CAGR (5.8%) and only 7.7% of the company's gain. **So what:** check for one-off orders before quoting a growth rate.



Insight 3 - Central's +2,072% is a small-base effect

Central made only \$540 profit in 2014 (0.5% margin). Moving to \$11,717 in 2015 is +2,072%, yet the dollar gain of \$11.2k is smaller than West's +\$19.8k (+82%) in 2017 and East's +\$13.1k (+65%) in 2017. The flag in the Excel sheet marks only this one case as SMALL BASE. **So what:** the percentage is not a sign of a great year; it is a sign of a very low starting point.



Insight 4 - Central grows sales but loses profit

In 2017 Central sales were flat (-0.2%) while profit fell 62% to \$7.6k and the margin dropped from 13.5% to 5.1%. Central gives the highest average discount of all regions (22-27%, against 10-16% in West and East). Texas and Illinois together lost about \$15.6k in 2017 on roughly \$67.8k of sales; without those two states Central would have made about \$23k. **So what:** review discounting and product mix in Texas and Illinois.

Insight 5 - Compare the same quarter, not the previous quarter

Every region peaks in Q4. Central sold \$33.4k in Q3-2016 and \$68.1k in Q4-2016, a +104% jump if you compare quarter on quarter. Against Q4-2015 (\$42.6k) the real growth is +59.7%. The sequential number mixes seasonality with growth, so I used same-quarter-last-year comparisons for the quarterly view. **So what:** pick the comparison period that removes the seasonal pattern.

6. Recommendations

- Keep investing in West and East; they deliver most of the absolute growth.
- For Central, fix profitability before pushing more volume: audit discounts and the Texas and Illinois accounts.
- Report growth with three things together: percentage, dollar change and CAGR. Add an outlier check for regions with a small number of very large orders (South).

7. Limitations

- Only four years of data, so CAGR rests on three growth steps; one unusual year moves it a lot.
- Superstore is a sample dataset; the findings describe this data and not a real company. I did not separate the effect of price from quantity.
- Regions are large groups of states; a region-level number can hide good and bad states, as Central shows.

8. Interview questions

Why can growth rate mislead? Because it depends on the starting value. A tiny base gives a huge percentage (Central profit +2,072% = only \$11k), a one-off order can distort the base (South 2014), and seasonality or a badly chosen period can create growth that is not real (Q3 to Q4). A percentage also says nothing about size, so I show dollar change and CAGR next to it.

What is a small-base effect? When the previous period value is very low, even a small absolute increase looks like a very large percentage growth. The percentage then says more about the low base than about performance. Example from this project: Central profit going from \$540 to \$11,717.

Appendix - key SQL (year-over-year with LAG)

```
WITH yearly AS (  
  SELECT region, order_year, SUM(sales) AS sales  
  FROM orders_clean GROUP BY region, order_year)  
SELECT region, order_year, ROUND(sales,2) AS sales,  
  ROUND(100.0*(sales - LAG(sales) OVER (PARTITION BY region ORDER BY order_year))  
  / LAG(sales) OVER (PARTITION BY region ORDER BY order_year), 2) AS sales_yoy_pct  
FROM yearly ORDER BY region, order_year;
```

All files (SQL script, Excel workbook with formulas, charts and this report) are in the GitHub repository for this project.